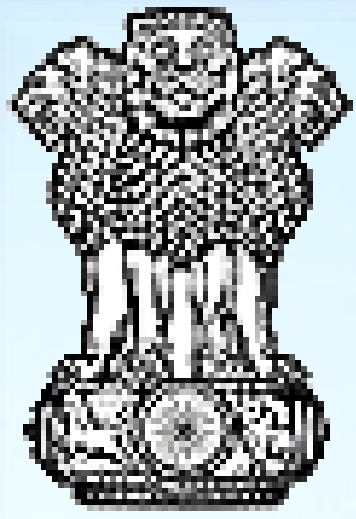




सत्यमेव जयते

Early warning System



M. Mohapatra

(m.mohapatra@imd.gov.in)

Director General of Meteorology

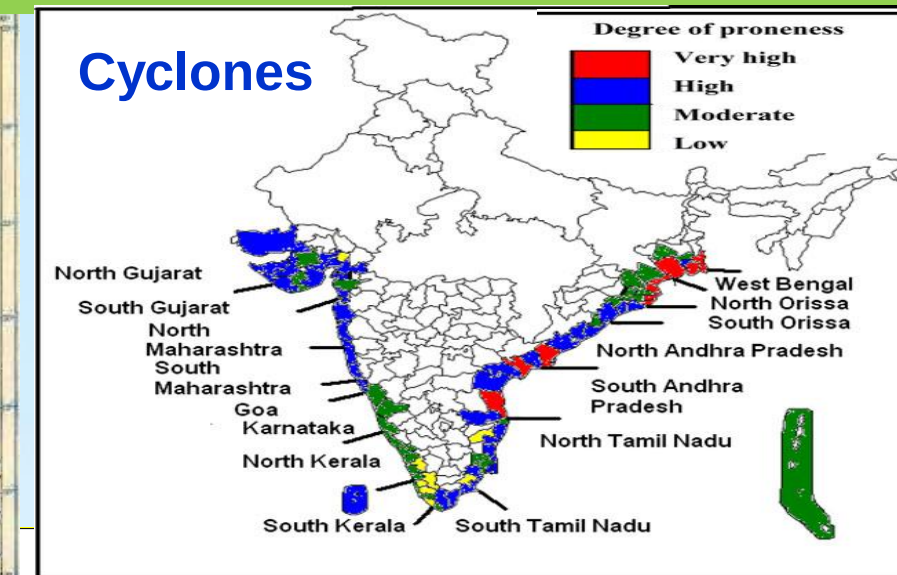
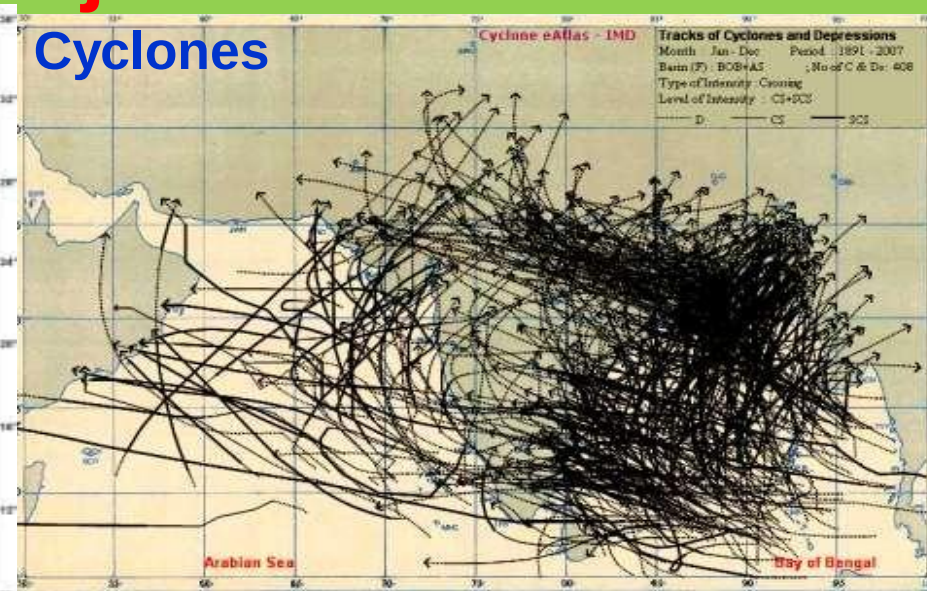
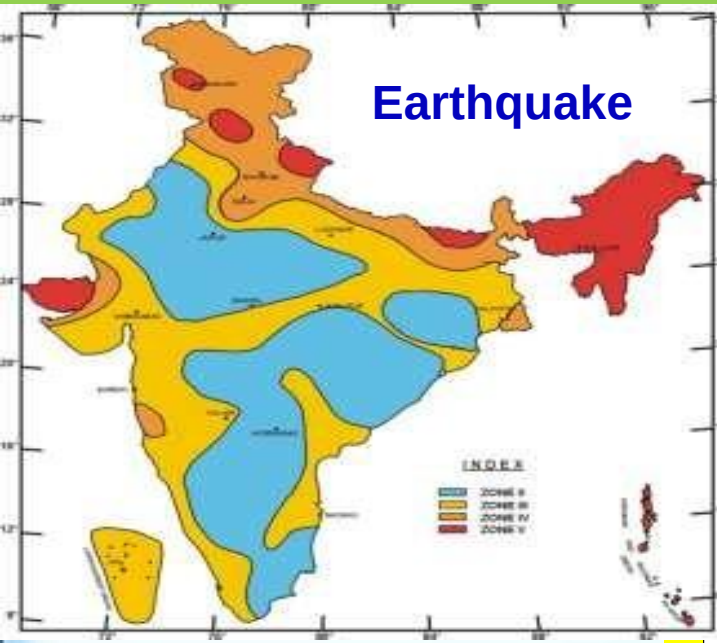
**India Meteorological Department , Ministry of Earth Sciences,
Government of India**



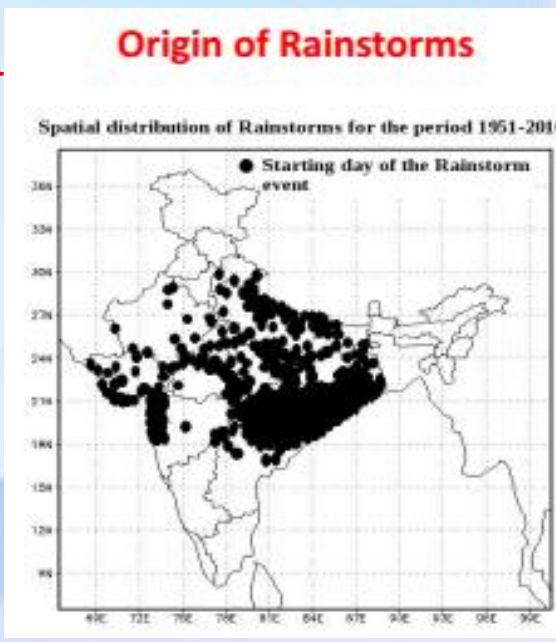
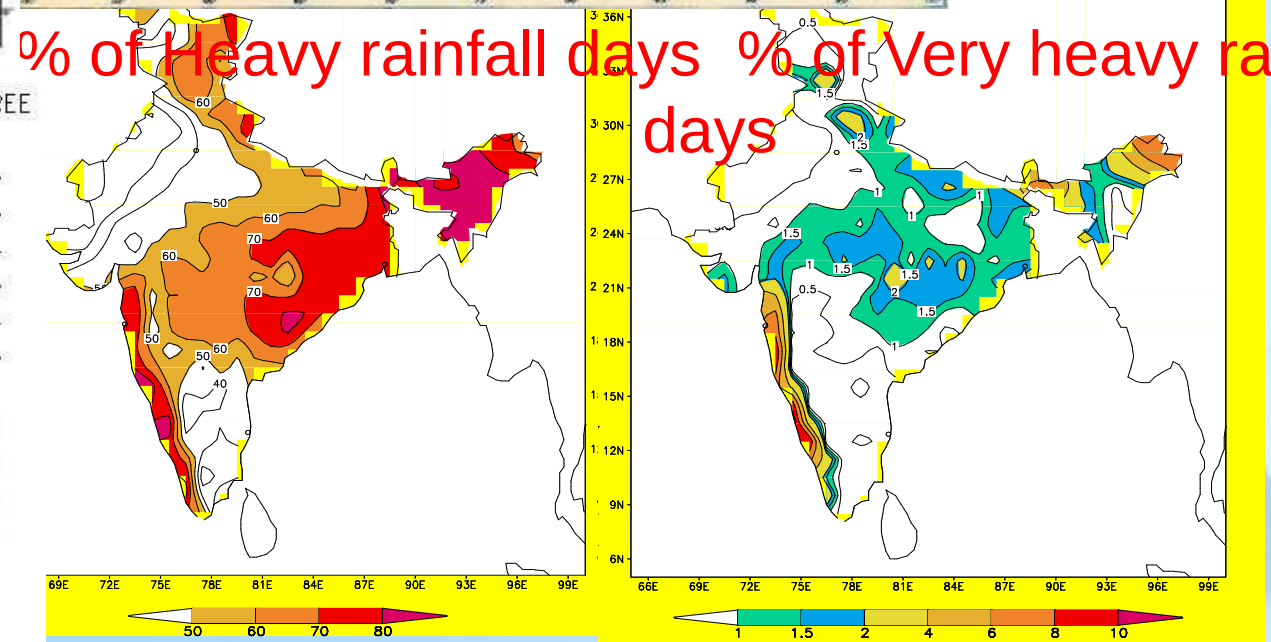
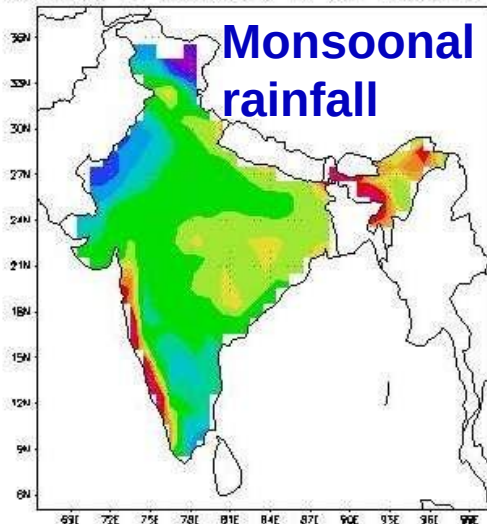
**भारत मौसम विज्ञान विभाग
INDIA METEOROLOGICAL DEPARTMENT**



Major natural hazards in India



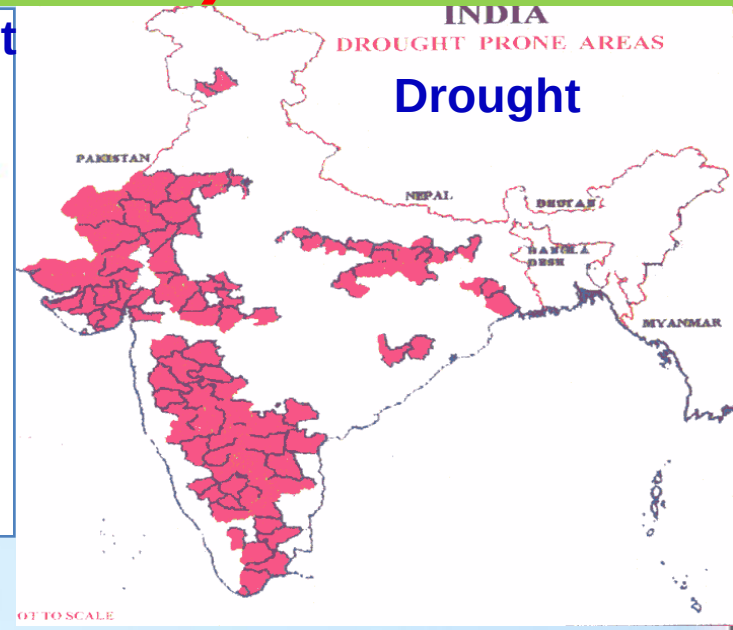
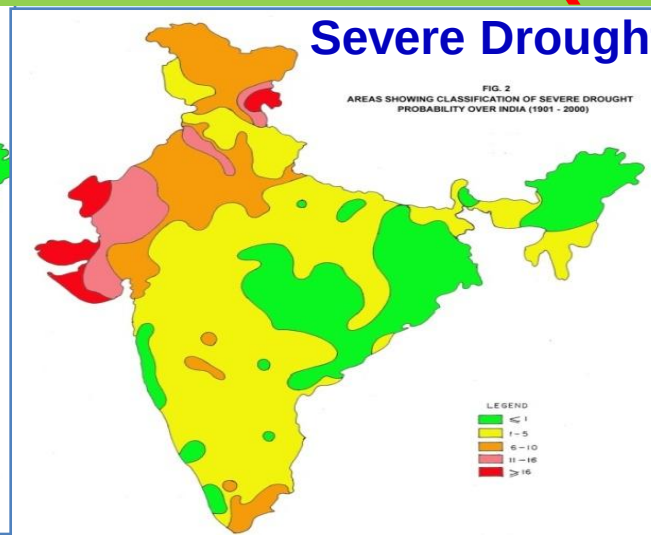
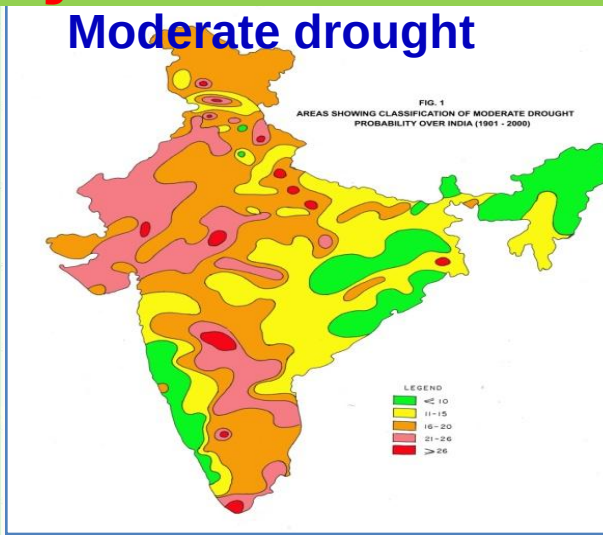
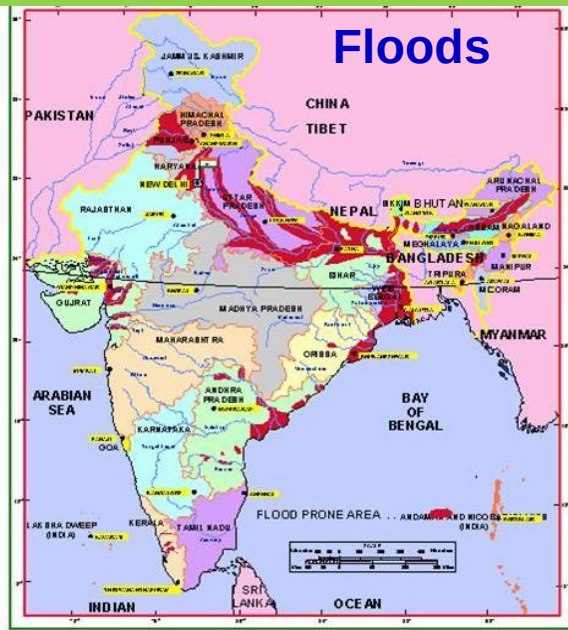
NORMAL RAINFALL IN CM (JUNE TO SEPTEMBER) 1.0 DEGREE



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INDIA METEOROLOGICAL DEPARTMENT

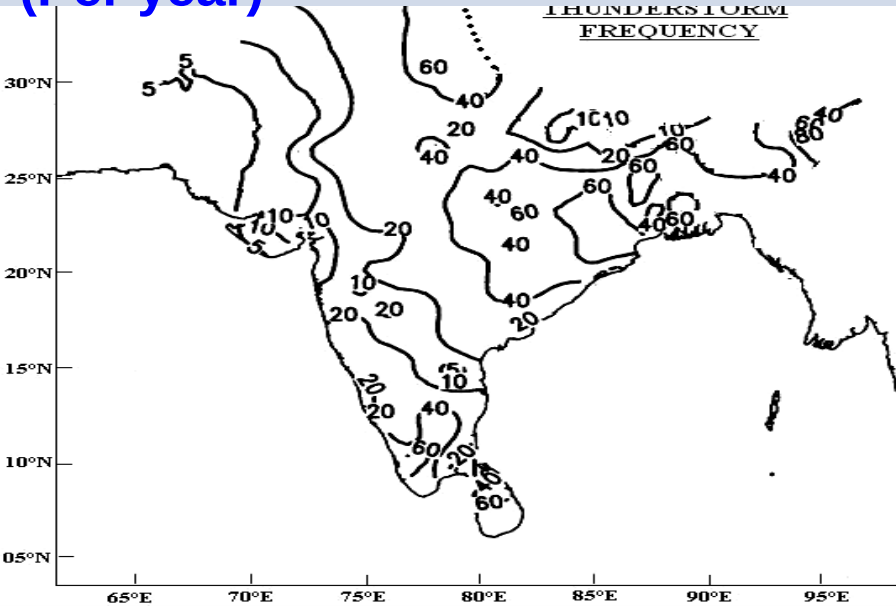


Major natural hazards in India (Contd.)

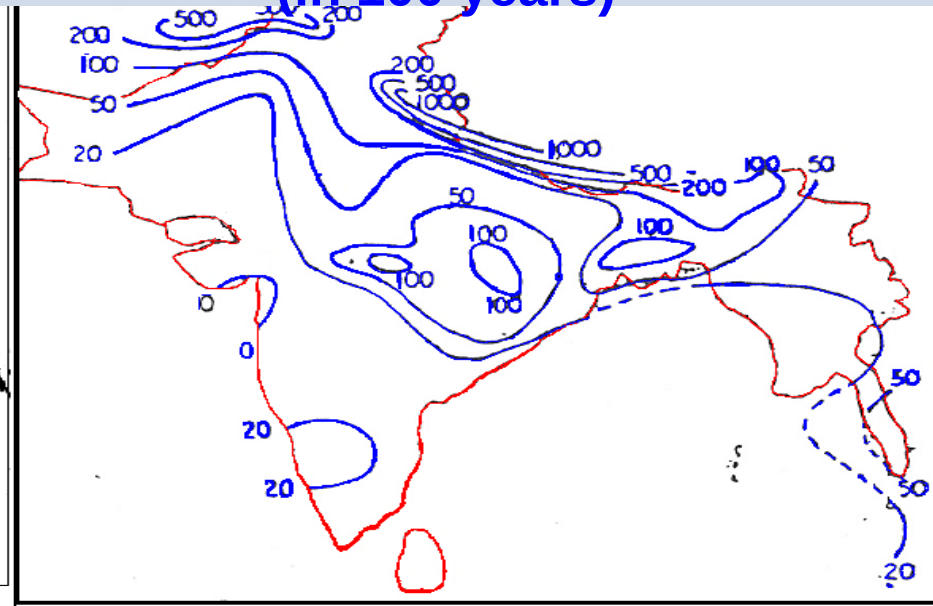


Meso-scale severe weather events

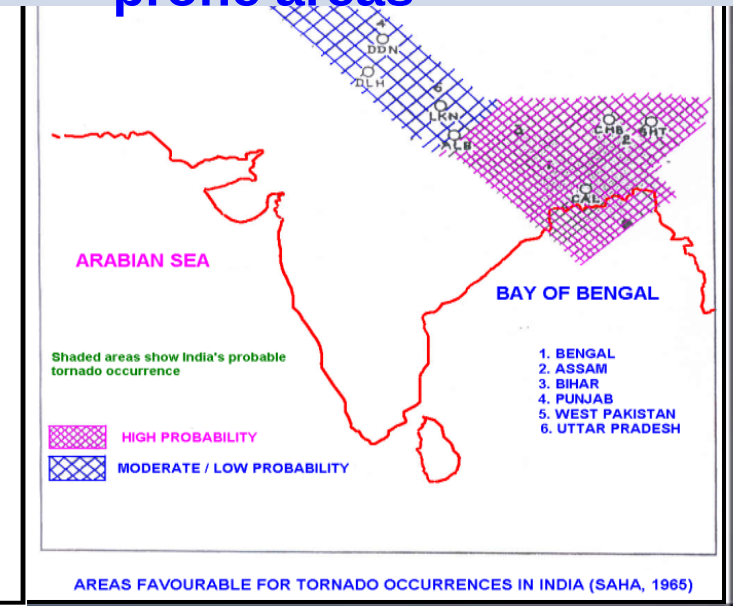
Thunderstorm (Per year)



Hailstorm (in 100 years)



Tornado prone areas



Institutional Arrangement in India for Multi-Hazard Early Warning System

HYDRO-METEOROLOGICAL HAZARDS –

IMD,

Tropical Cyclones, Local Severe Storms, Winter Systems.

CWC: Floods

[Support from IMD for heavy rain and quantitative precipitation forecasting),

Ministry of Agriculture: Drought

[Support from IMD for weather monitoring and forecasting)

SASE, DRDO : Snow Avalanches

[Support from IMD for heavy snowfall warning)

MoEFCC: Climate change

(Climate change impacts on severe weather events (IITM and IMD)

GEOLOGICAL HAZARDS

Earthquakes & Tsunamis (NCS and INCOIS)

[Support for Rain Induced Landslides/Mudslides (IMD)]

ENVIRONMENTAL IMPACTS

-Air pollution & Haze, FOG, Smog (IMD)

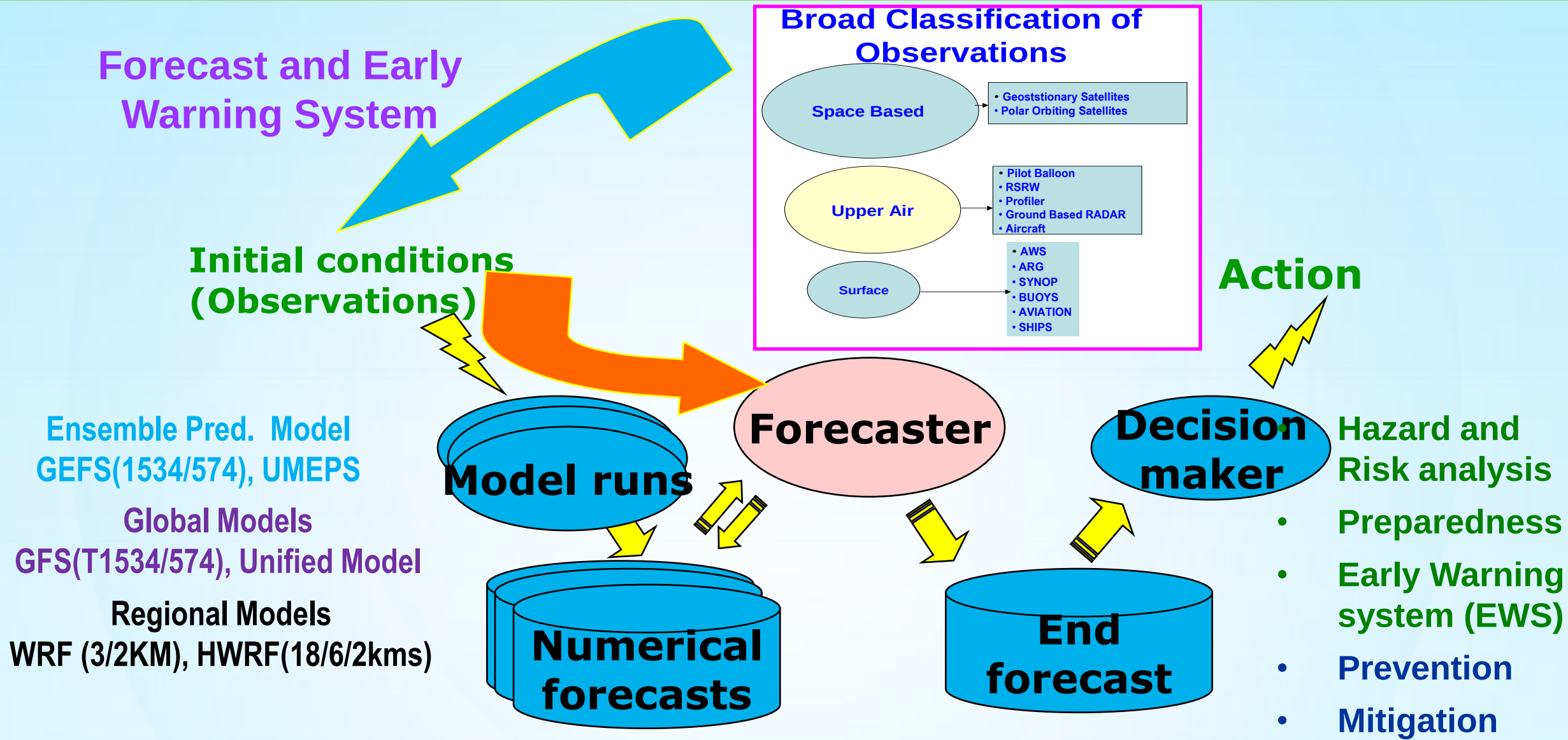
-Coastal Zone Management (NCCR)

-Coastal Erosion (NCCR)

-Eco-system monitoring/ modeling (IITM and IMD)



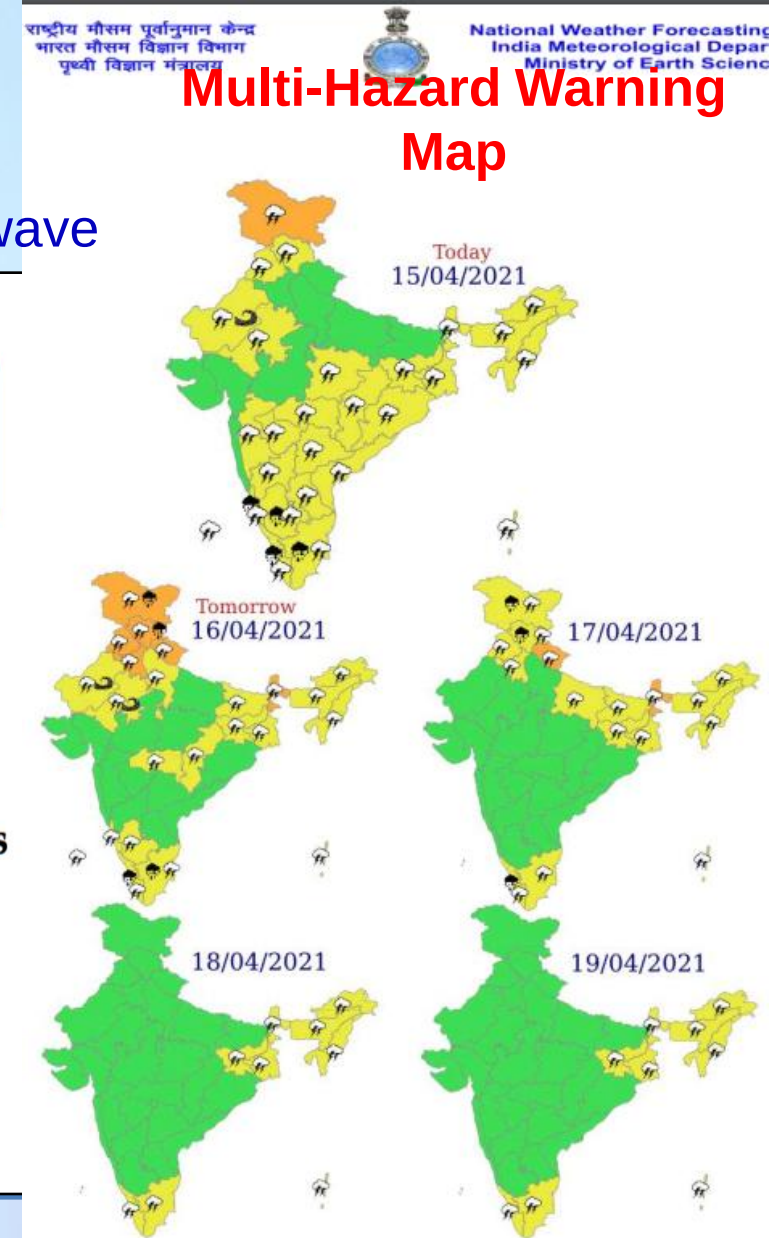
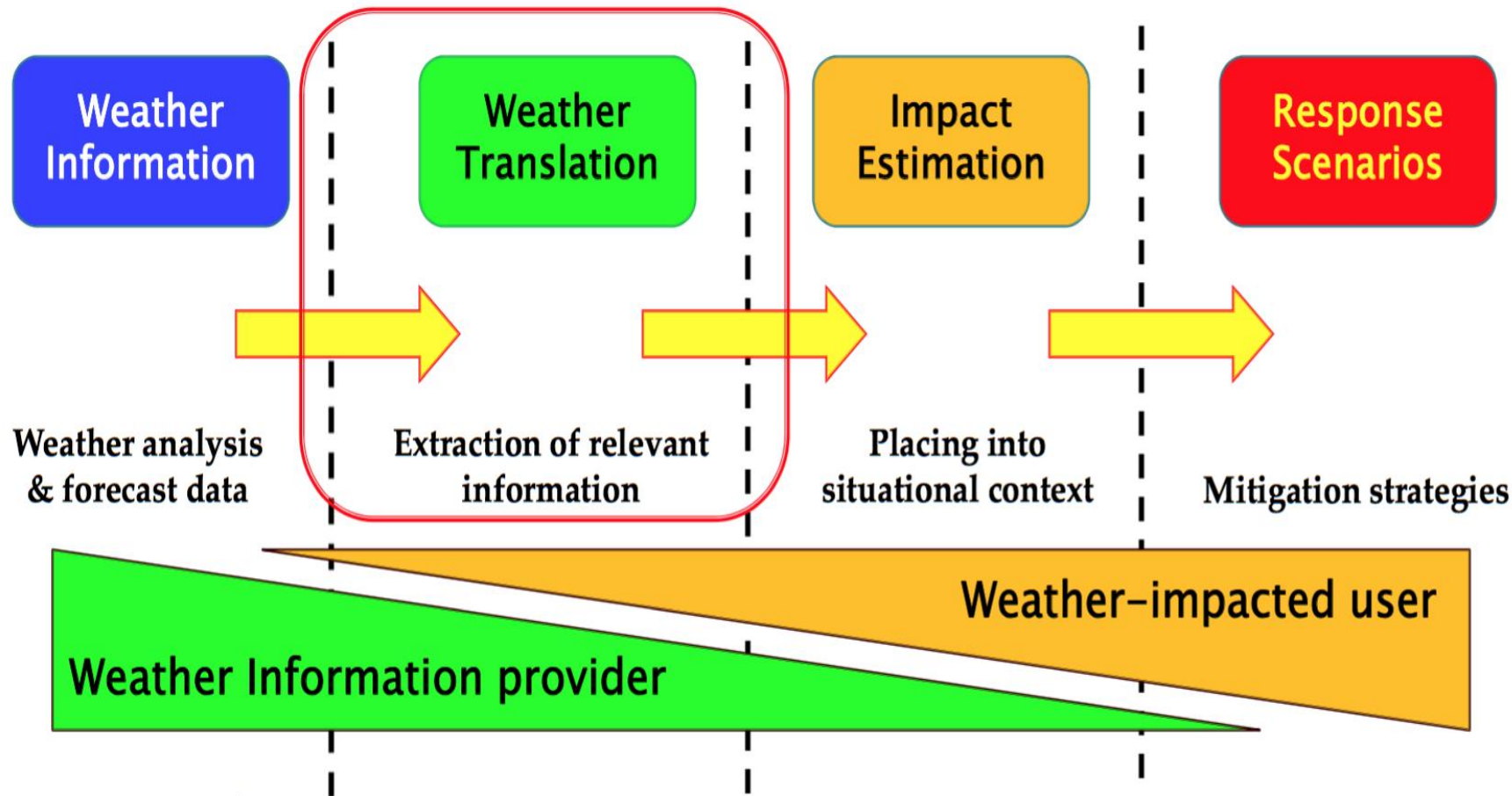
Schematic Multi-Hazard Early warning System in IMD



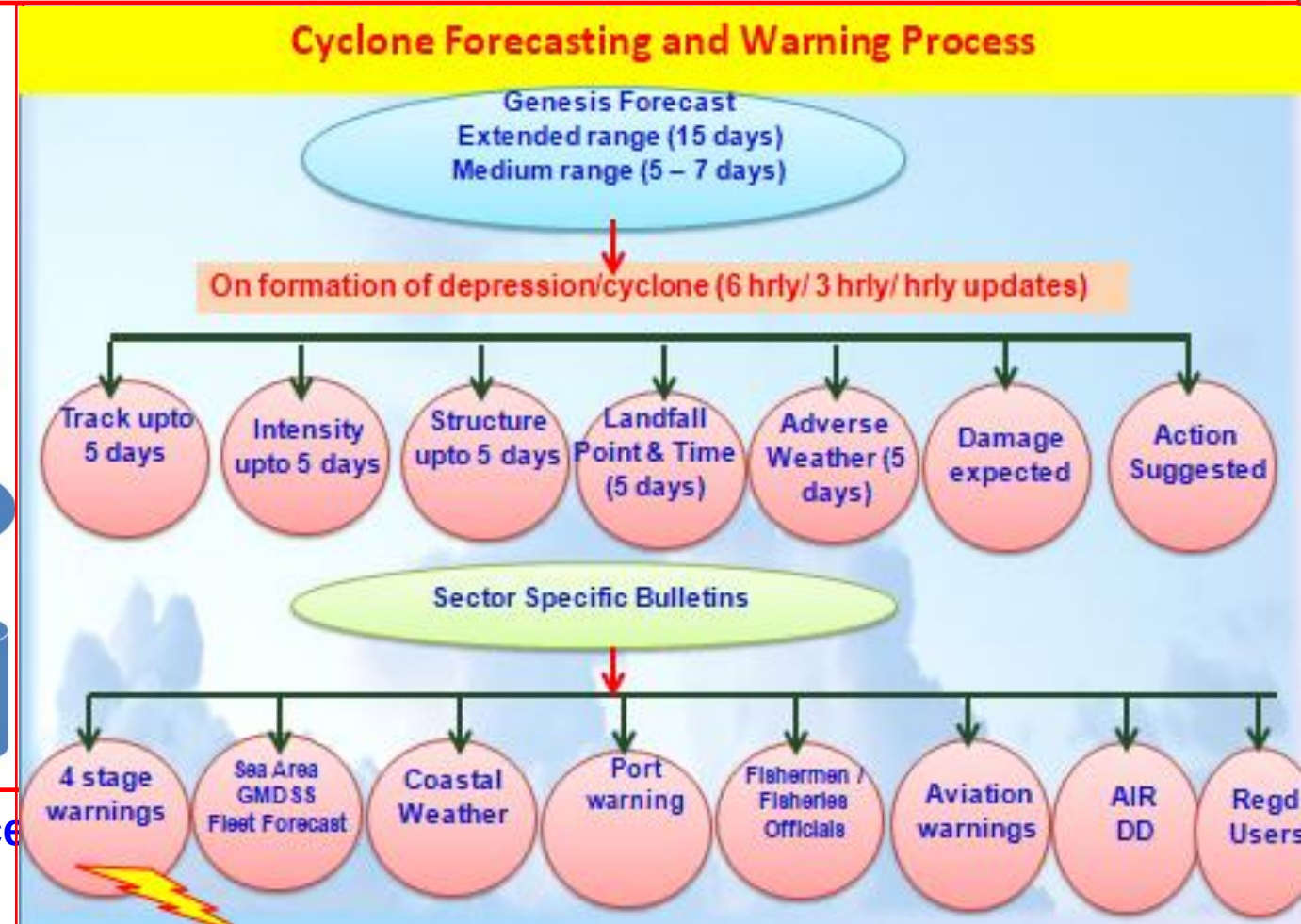
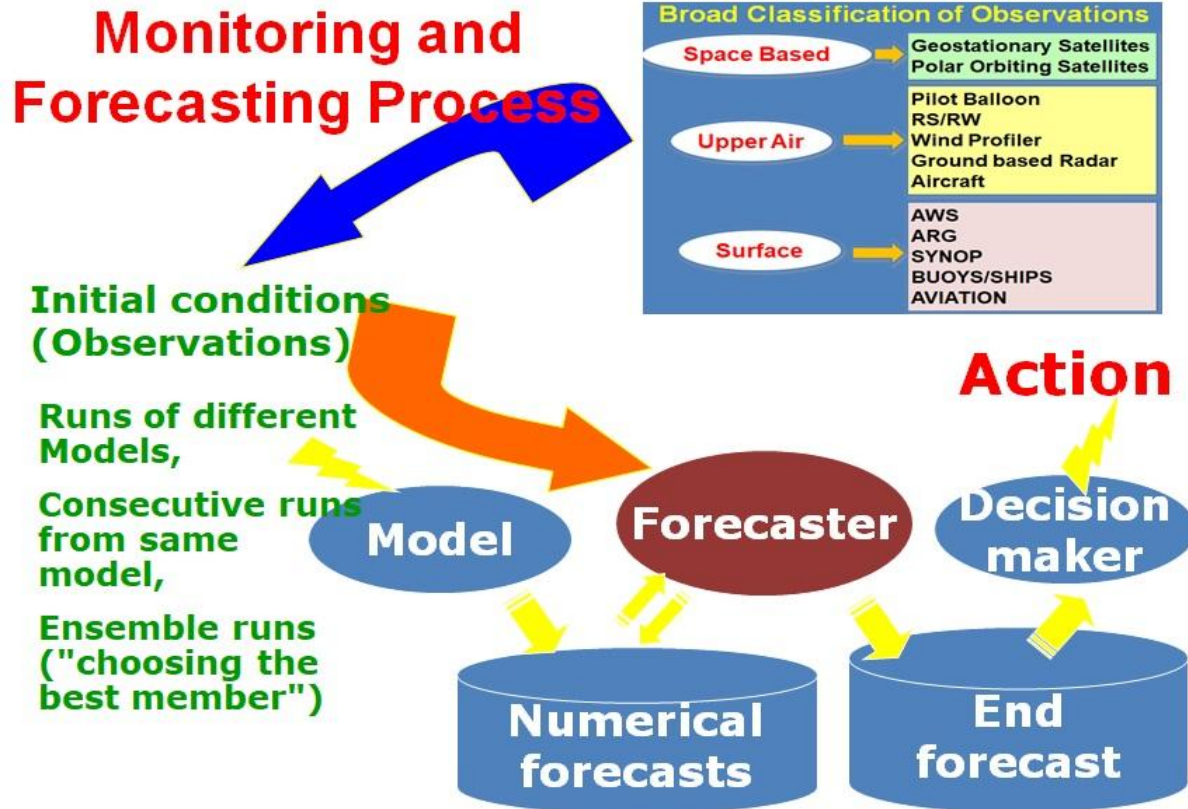
Improved Forecast and Early warning system with respect to all the above components

Impact-based Forecasting in India

- Relevant information from weather forecast is extracted and placed into situation context to produce impact estimations;
- With potential impact information, response scenarios are set-up
- Targeted Hazards: Cyclone, Heavy rain/flood, Thunderstorm, Heat/cold wave



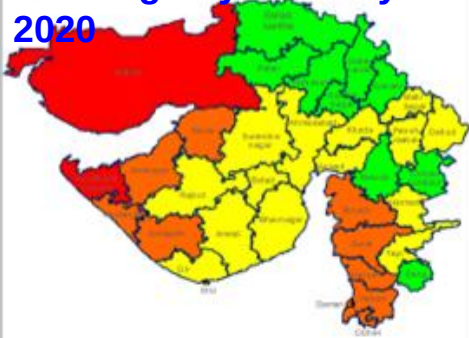
Cyclone Monitoring and Forecasting Process of IMD



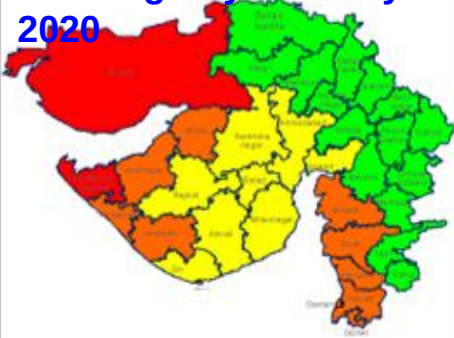
- ❖ Pre-cyclone watch (Yellow) –72 hours in advance
- ❖ Cyclone Alert (Orange)- 48 hrs in advance.
- ❖ Cyclone warning (Red) – Issued at least 24 hrs in advance indicating latest position of Tropical Cyclone, intensity, time and point of landfall, storm surge height, type of damages expected and actions suggested.
- ❖ Post-Landfall Outlook- Issued about 12 hrs before landfall & till cyclone force winds prevail; District Collectors of interior districts besides the coastal areas are also informed.
- ❖ 'De-Warning' message issued when the Tropical Cyclone weakens into Depression stage.

Impact Based Forecast(IBF) and Warnings for Heavy Rain

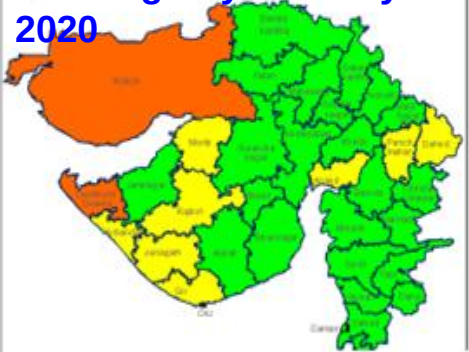
Warning Day 1: 5 July 2020



Warning Day 2: 6 July 2020



Warning Day 3: 7 July 2020



Warning Day 4: 8 July 2020



Warning Day 5: 9 July 2020



Warning(Take Action)

Alert(Be Prepared)

Watch (Be Updated)

No Warning (No Action)

IMPACT BASED FORECAST

Expected Impact with respect to red colour warning issued for the Districts namely Devbhoomi Dwarka, Porbandar,

Jamnagar, Kutch:

- ✓ Major damage to kuchcha roads due to inundation.
- ✓ Major disruption in traffic in city areas.
- ✓ Inundation of low lying areas leading to damage to kuchcha houses.
- ✓ Water logging in underpass in city areas.
- ✓ Sudden reduction in visibility during heavy downpour leading to road accidents.

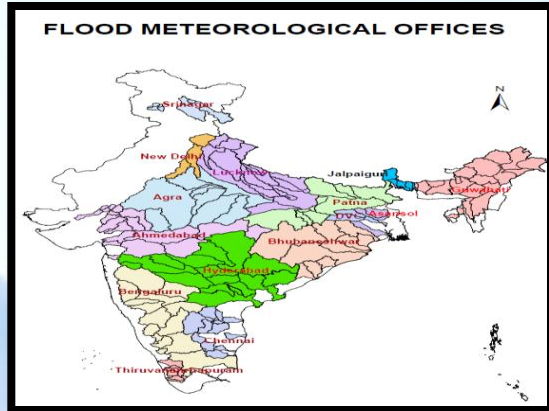
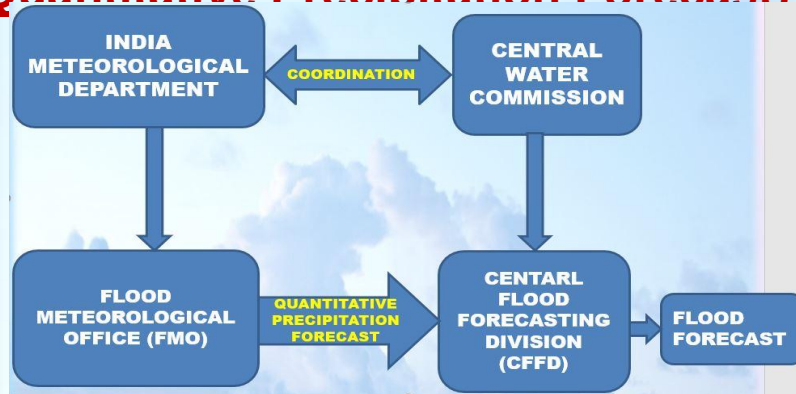
IBF & Warning Stages

- **Stage -1:** Heavy rainfall Watch-(3-4 days lead time daily update)
- **Stage-2:** Heavy rainfall Alert: (48 hours prior to the occurrence of event at 12 hourly updates)
- **Stage-3:** Heavy rainfall Warning (24 hours prior to the occurrence of event at 06/12-hourly updates)
- **Stage-4:** 12-Hours prior to occurrence event-at 3-hourly updates



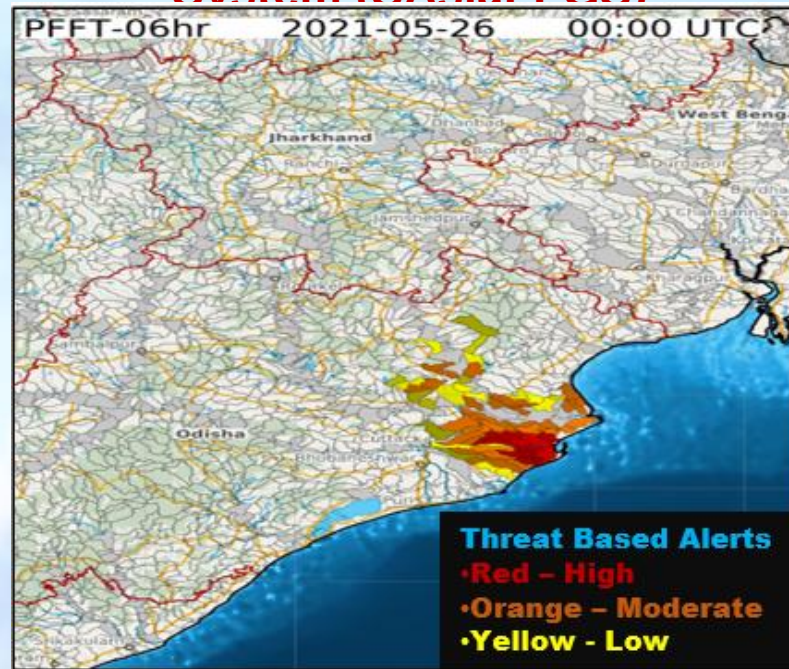
Hydromet Services of IMD for Flood Forecasting

Riverine Flood Forecasting (Quantitative Precipitation Forecast)



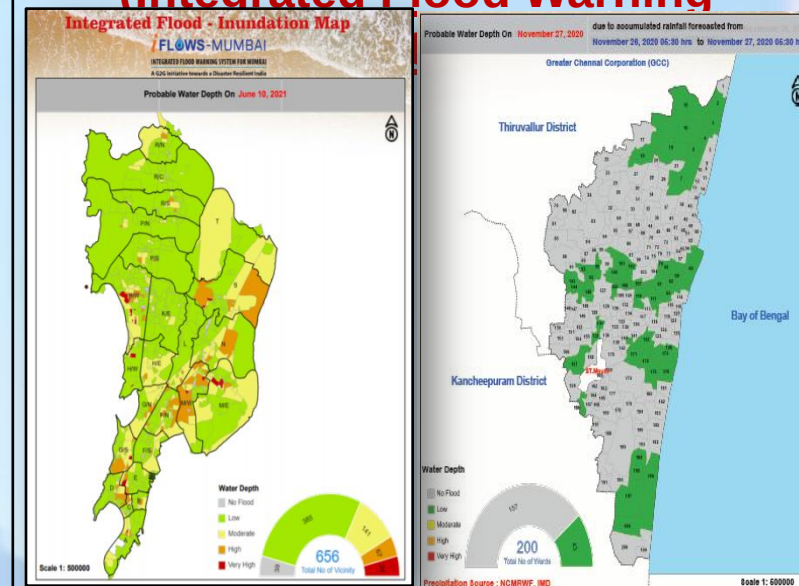
- Hydromet services through 15 Flood Met. Offices for 153 river sub basins.
- Sub-basin-wise Quantitative Precipitation Forecast: and Heavy rainfall warnings for day:1-5.

South Asia Flash Flood Guidance System (SAsiaFFGS)



- First of its kind in South Asia (Bangladesh, Bhutan, India, Nepal, Sri Lanka).
- High resolution (4X4 km) and 30000 watersheds over Indian region.
- Capable of issuing flash flood Threat and risk for next 6 and 24 hours respectively.

City Specific Flood Forecast (Integrated Flood Warning)



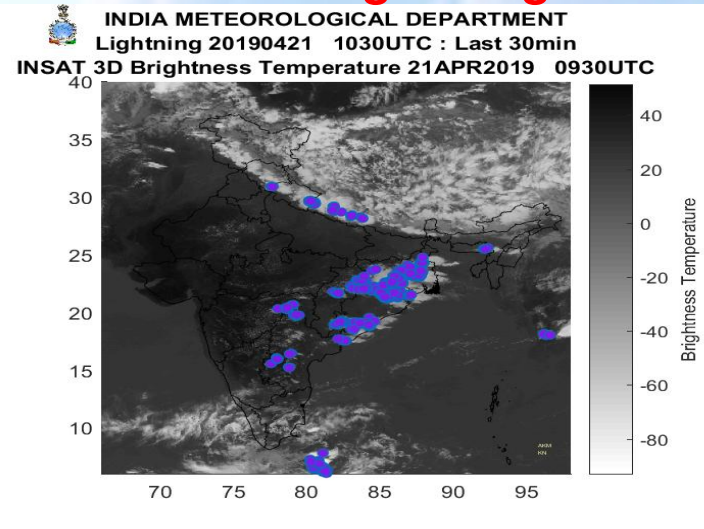
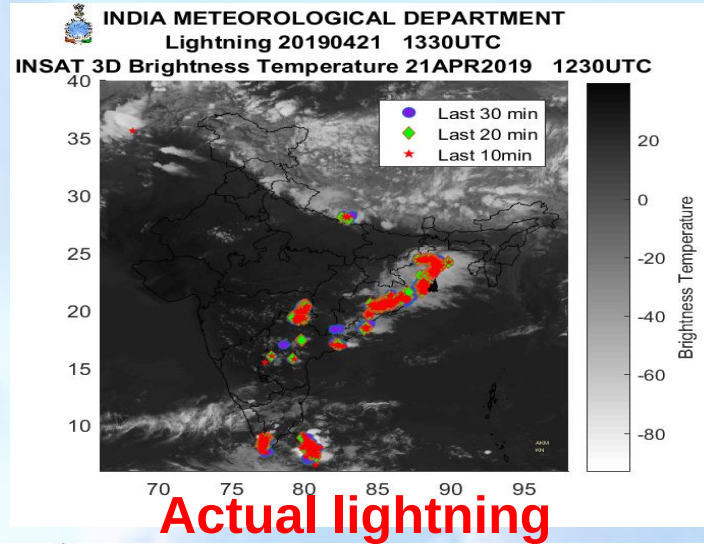
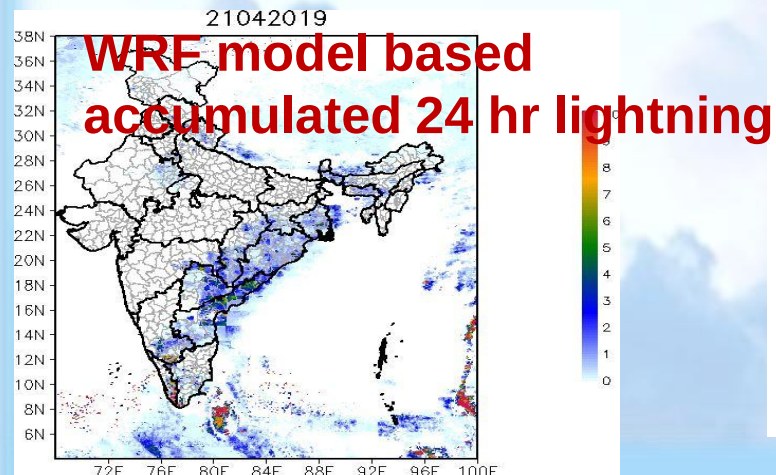
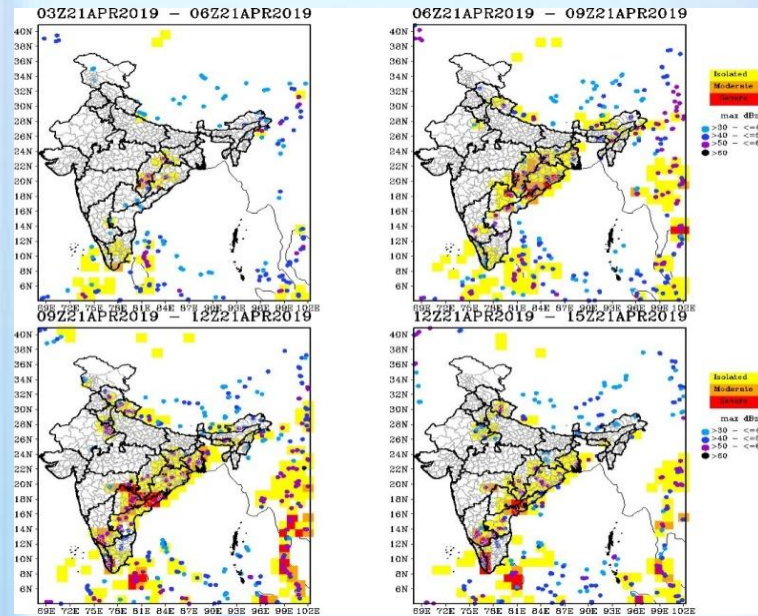
Code	Water Depth (feet)
	No flood
	3-4
	4-5
	5-6
	>6

- Implemented at Mumbai and Chennai City.
- Ward-wise probable water depth/inundation.



Thunderstorm Early Warning System

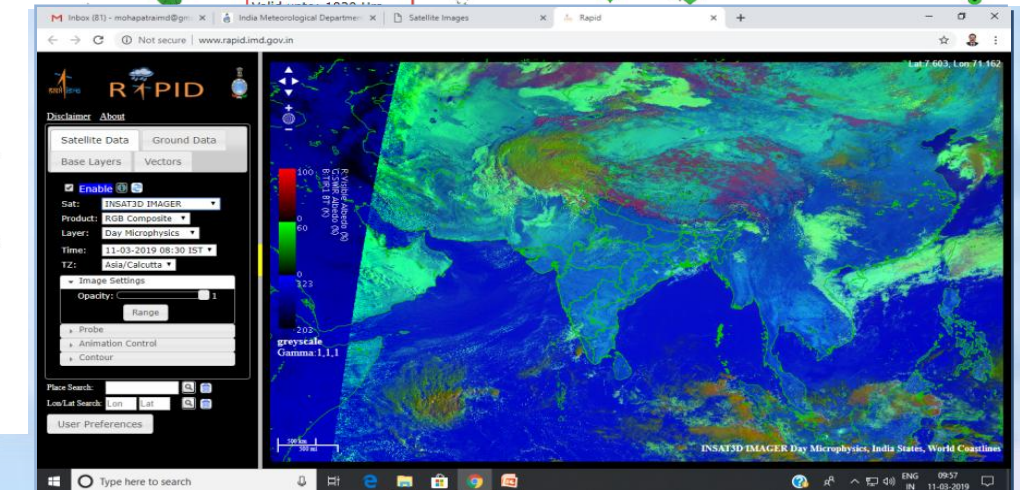
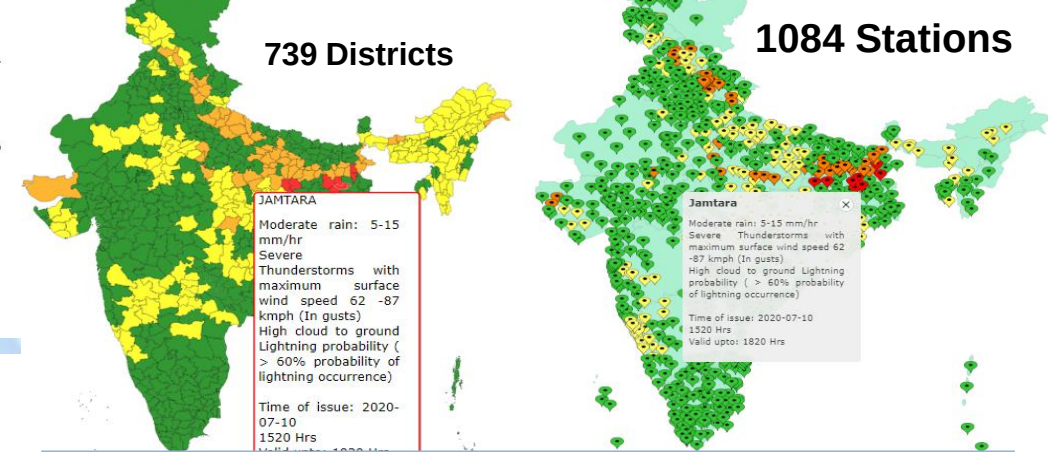
WRF model based 3 hourly lightning forecast



Models:

GFS, NCUM, WRF, NCUM-R, GEFS, UM-EPS, ECMWF, MeteoFrance, JMA, NCEP-GF

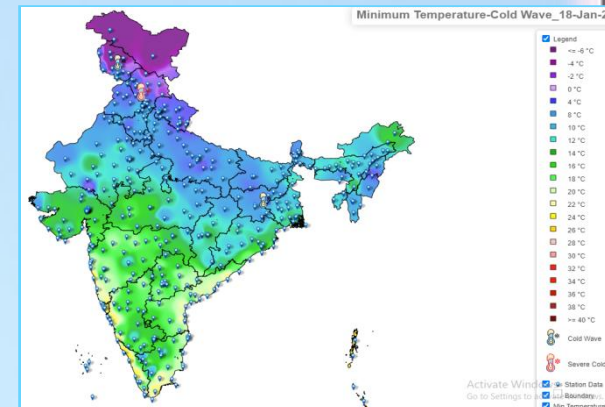
Three hourly thunderstorm warning



Heat/cold Wave disaster management

Heat/cold wave warning in Web-GIS

- Seasonal and extended range (upto two weeks) outlook
- District level heat/cold wave warning (upto five days)
- Heat action plan with different states
- Heat action plan for cities
- Daily two times (0830 & 1600 hours IST) impact based heat wave warning

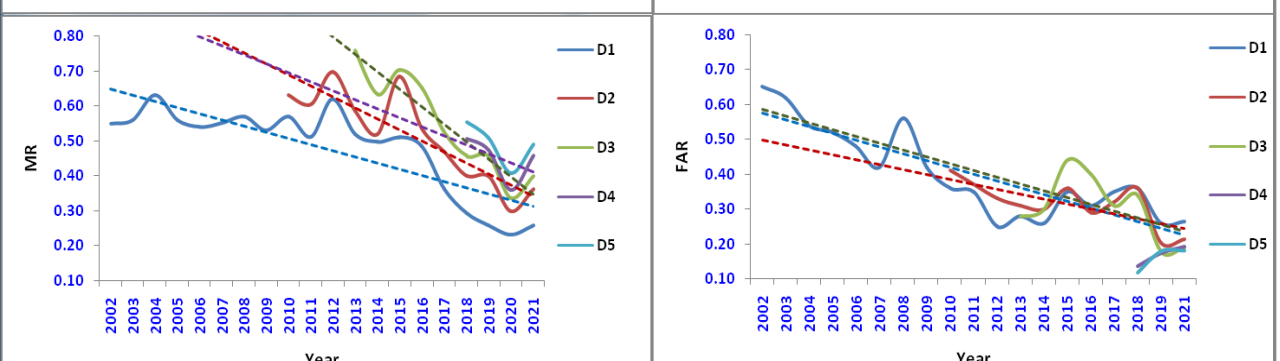
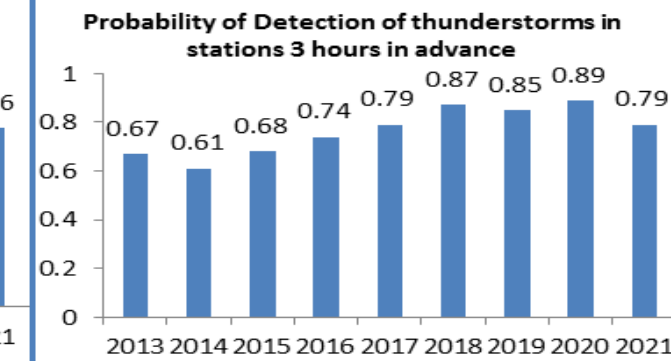
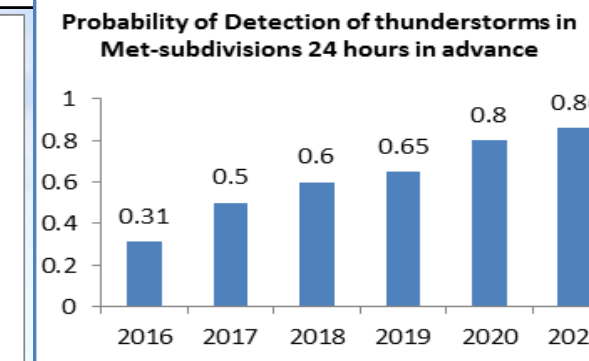
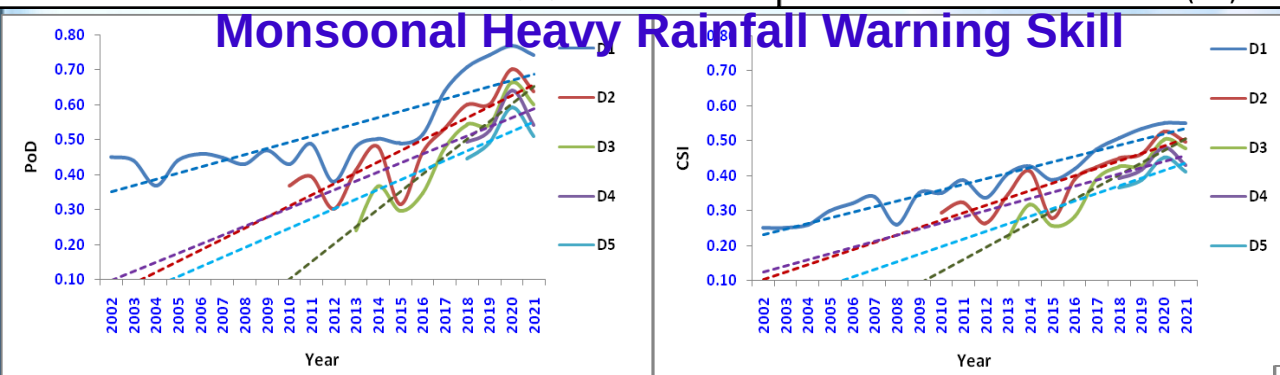
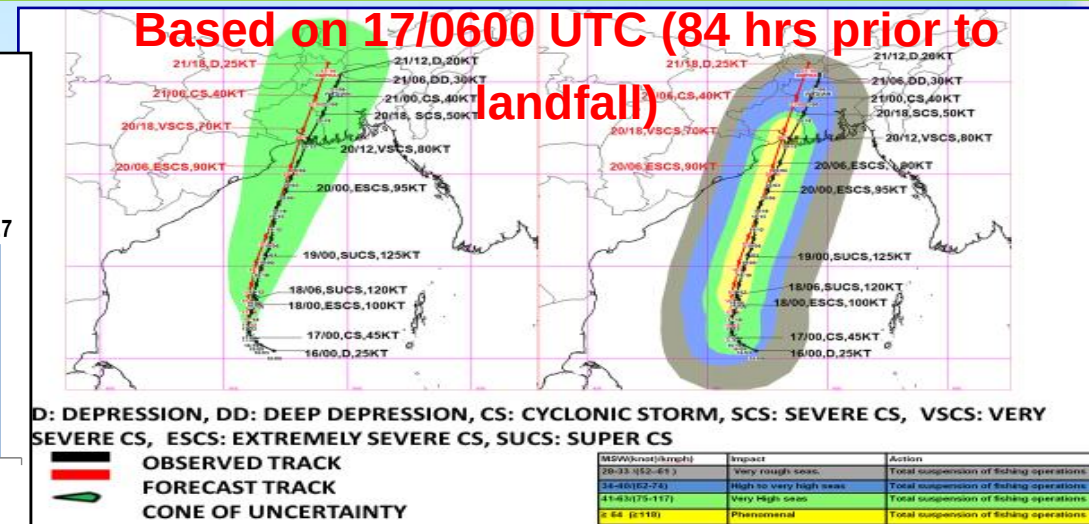
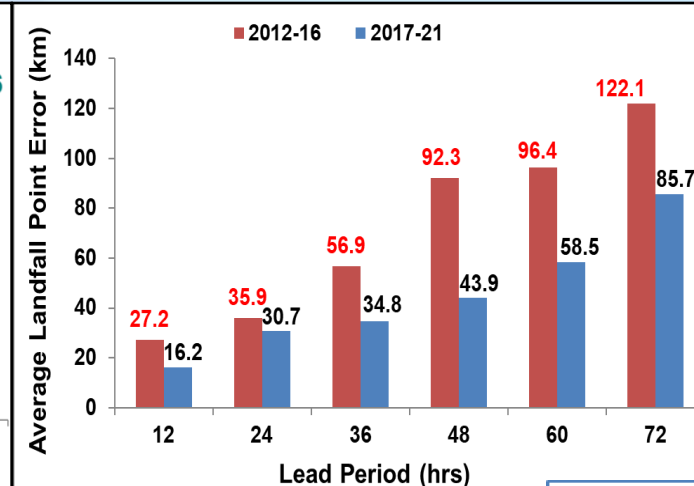
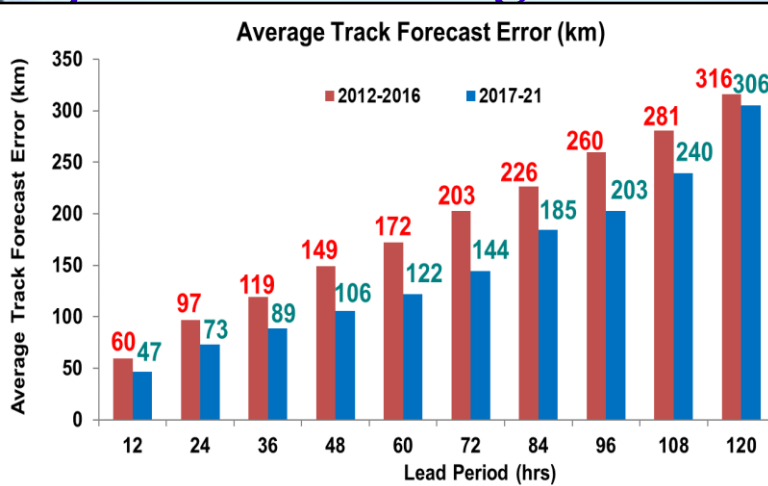


Warning	Impact	Suggested Actions
Nil	Comfortable temperatures	No cautionary action required
Heat wave conditions at district level, likely to persist for 2 days	Heat is tolerable for general public but moderate health concern for vulnerable people e.g. infants, elderly, people with chronic diseases.	Avoid heat exposure
i. Severe heat wave conditions likely to persist for 2 days. ii. With varied severity, heat wave is likely to persist for 4 days or more.	Increased likelihood of heat illness symptoms in people who are either exposed to sun for a prolonged period or doing heavy work. High health concern for vulnerable people e.g. infants, elderly, people with chronic diseases.	Avoid heat exposure— keep cool. Avoid dehydration
i. Severe heat wave likely to persist for more than 2 days. ii. Total number of heat/severe heat wave days likely to exceed 6 days.	Very high likelihood of developing heat illness and heat stroke in all ages.	Extreme care needed for vulnerable people.



Forecasting and warning skill of IMD

Cyclone Warning Skill



40% improvement in accuracy in recent five years compared to previous five years Skill

Warning Dissemination system

- ❖ Telephone, Tele-fax, Mobile Phones (SMS) through IMD severe weather network, Agromet Network, INCOIS network.
- ❖ VHF/HFRT/Police Wireless, Aeronautical Fixed Terminal Network
- ❖ Global telecommunication system (GTS) :
- ❖ NAVTEX , Internet (e-mail), ftp
- ❖ Mass Media: : Radio/TV, News Paper network (AM, FM, Community Radio, Private TV) : Prasar Bharati and private broadcasters, Websites, Dedicated websites and web pages, Social media, Weekly and daily Weather Video
- ❖ GAMES and NAVIK



Damini for lightning



Meghdoot for
Agromet services



Mausam for
weather
information and
warnings

- » Public Website (mausam.imd.gov.in)
- » IMD Apps: Mausam/ Meghdoot/DAMIN/RAIN ALARM, UMANG
- » Social Media: Facebook, Twitter, Instagram, BLOG
- » Twitter: <https://twitter.com/Indiametdept>
- » Facebook::
- » <https://www.facebook.com/India.Meteorological.Department/>
- » Blog: <https://imdweather1875.wordpress.com/>
- » Instagram:https://www.instagram.com/mausam_nwfc
- » Youtube:https://www.youtube.com/channel/UC_qxTReoq07UVARm87CuyQw



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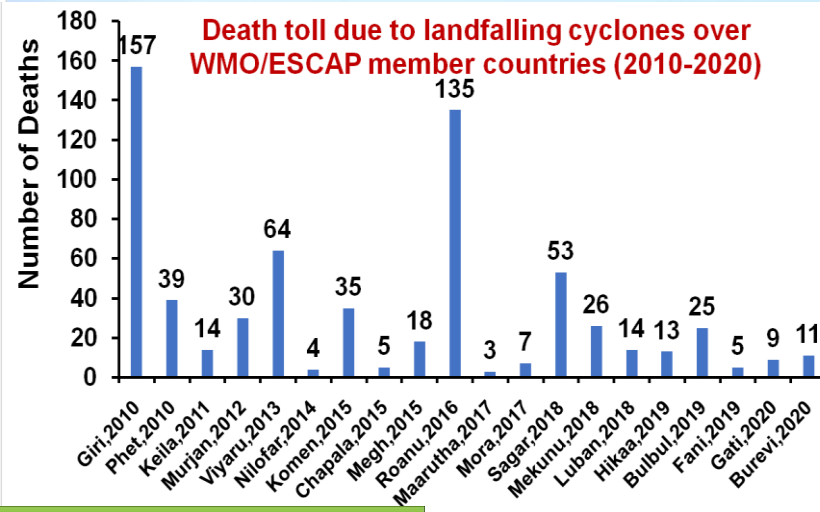
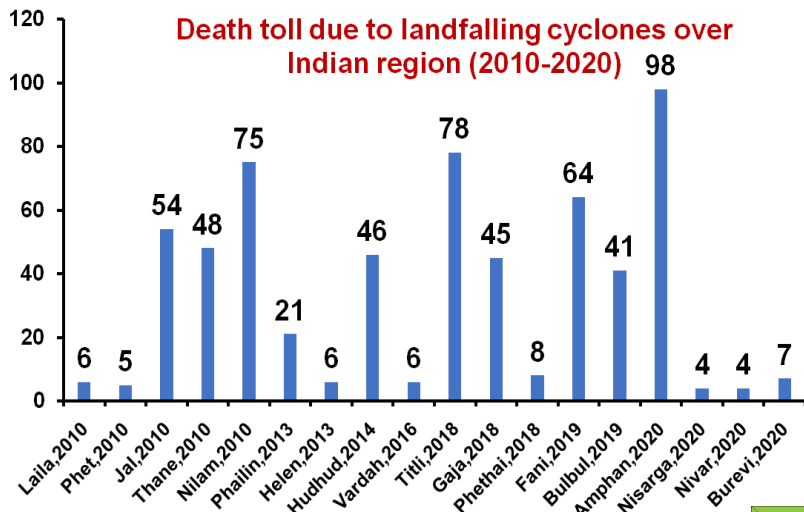
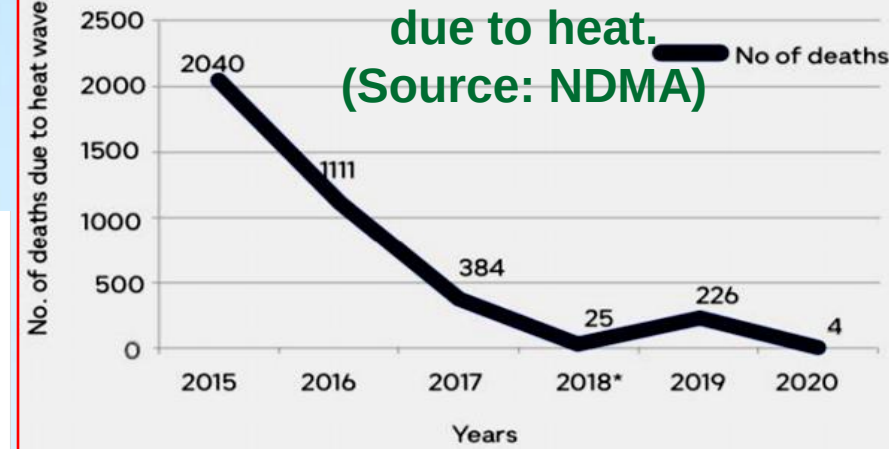


Outcome of Improved Early Warning System

- Improved confidence of disaster managers/general Public
- Enabled disaster managers and public to minimize loss of lives and properties
- Brought laurels to the country from international community

Reduction in number of deaths due to heat.

(Source: NDMA)



FANI, 2019

Our expertise in accurate weather forecast has improved. This was evidenced during the recent #CycloneFani that struck the eastern coast of the country. Due to accurate information and timely preparation, large scale destruction to life and property was averted



HUDHUD, 2014

In this cyclone, India Meteorological Department made excellent utilisation of Technology and from 6th October itself, this cyclone was predicted. The actual wind speed due to cyclone was same as the predicted wind speed. The track of the cyclone was same as that predicted. The time of landfall of cyclone was also same as that predicted by IMD.



WMO Appreciation: Mekunu, 2018

Subject: Severe Tropical Cyclone "Sagar" and Extreme Severe Tropical Cyclone "Mekunu"

Dear Dr. Ram Nath Kovind,

I wish to refer to the recent severe tropical cyclone "Sagar" and the extreme severe tropical cyclone "Mekunu", which happened consecutively in the Arabian Sea, from 17 to 28 May 2018.

I wish to express my sincere gratitude and appreciation to the ISM Tropical Cyclone Centre, New Delhi, India that provided its advisory bulletins every three hours, during these two events.

The advisory bulletins were well distributed to all Members of the WMO/ESCAP Panel on Tropical Cyclones (PTC) and Somalia, for their early preparedness. They were also distributed to WMO Secretariat in Geneva, and forwarded to WMO Regional Offices in Bahrain and New York.

In addition to benefiting PTC Members and Somalia, these bulletins were well utilized by WHO Regional Office in Bahrain, to communicate and coordinate with both PTC and other Members neighboring Oman for their early preparation and necessary actions in response to the tropical cyclones. At the same time, all members of the Inter-Agency Standing Committee (IASC) with the key UN and non-UN humanitarian partners were informed and WMO representative in New York used the bulletins to provide a daily briefing to the United Nations (UN) Operations and Crisis Centre at UN Headquarters, and informed the UN Secretary-General's Executive Office on the development of the tropical cyclones and their potential impacts.

It is vitally important and effective practice to distribute and communicate ISMDC Dewey Delhi's advisory bulletins with Members, WMO, UN and its humanitarian agencies in a timely manner, to enable all those concerned to take necessary and appropriate actions. I shall appreciate it very much if such a practice will be continued and strengthened.

I look forward to your continued support to the WMO Tropical Cyclone Programme and activities to reduce risks of disasters by tropical cyclones.

Yours sincerely,

(P. Talasila)
Secretary-General

UN Appreciation: FANI, 2019

UN agency for disaster reduction has commended the Indian Meteorological Department's "almost pinpoint accuracy" of early warnings that helped authorities conduct a well-targeted evacuation plan and minimise the loss of life.

NEWS18 • 3 MIN READ

UN Praises Met Dept for 'Pinpoint Accuracy' on Cyclone Fani That Minimised Loss of Life

Phailin, 2013

World Meteorological Organization

On behalf of the World Meteorological Organization and myself, I wish to convey my heartfelt condolence and profound sympathy to you and your family in the circumstances of the untimely passing of your son, Dr. Phailin.

In the meantime, I also wish to commend your determination to deliver high quality forecast information and warnings to the government authorities and the public in the circumstances of already published super cyclone intensity of "Phailin" by the media at the time.

In this regard, I am convinced that the WMO's collective priority to reduce risks and potential impacts of hazards caused by weather, climate, water and related environmental elements, involve WMO's capabilities for service delivery at local, national and regional levels, will ensure that the role of the WMO is well reflected in national disaster risk reduction policies. With the advent of sub-seasonal and seasonal forecasting capabilities, I hope that through the use of our high quality forecast information, we will be able to deliver more targeted and timely disaster preparedness measures, thus, under the Global Framework for Climate Services, we would have the opportunity to further expand the contribution of our community to improving livelihoods and reducing further loss of lives associated with extreme weather, water and climate events.

Again, I wish to congratulate your service for providing timely and accurate forecast of cyclone track and intensity, which led to early evacuation and reduced human and property losses. I look forward to the continued strong working relationship between India and WMO and I wish to convey my sincere appreciation to you as well as my own to assist your Service and your Government in addressing disaster risk reduction challenges.

Yours sincerely,

(P. Talasila)
Secretary-General

ICHL AWARD, PHAILIN 2013

Indian Institute of Management Raipur

ICHL 2013 Award for Excellence in Humanitarian Action

Indian Meteorological Department

Government of India

in the field of

Early Warning, Forecasting and Dissemination

on 3rd Day of December, 2013

for the precise way in which their tracking of cyclone Phailin from 8th October 2013 onwards, the forecasting of the cyclone trajectory and the landfall location helped in the evacuation of more than one lakh people and their safe movement to cyclone shelters in Odisha and in Andhra Pradesh and minimised the loss of lives due to Cyclone Phailin in the affected areas.

The Indian Meteorological Department was able to ensure the minimum loss of lives in spite of several severe reports globally predicting otherwise, comparing the wind speeds and wind velocity to the 1999 super cyclone, mainly because of the installation of Doppler weather radars, application of state-of-the-art weather monitoring software models, modernization of IMD monitoring network and the absence of the IMD technical staff.

Dr. P. Talasila
Secretary-General

THANKS



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INDIA METEOROLOGICAL DEPARTMENT

